

PAKOWKI LAKE AGCM, AB, Canada

WMO: 710700

Lat: 49.2247N

Lon: 111.1261W

Elev: 915

StdP: 90.81

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.4	-25.9	-32.7	0.2	-29.3	-29.0	0.3	-25.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	32.6	15.7	30.7	15.7	28.6	15.3	18.5	26.1	17.4	26.1	16.4	25.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	12.6	21.5	14.5	11.5	19.7	13.4	10.7	18.6	56.1	25.3	52.4	26.0	49.4	25.4	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-34.4	35.9	2.5	1.7	-36.2	37.1	-37.6	38.1	-39.0	39.0	-40.8	40.2		
(5)			-34.5	20.3	2.4	1.7	-36.3	21.6	-37.7	22.6	-39.1	23.6	-40.9	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-7.2	-8.0	-0.8	5.3	11.1	15.5	19.1	18.5	13.6	6.1	-1.0	-8.0		
	DBStd	11.63	8.34	8.80	7.64	5.10	4.43	3.40	2.97	3.56	4.84	5.42	7.70	8.80		
	HDD10.0	2620	533	503	336	152	41	3	0	1	19	142	333	557		
	HDD18.3	4837	791	736	593	391	226	98	27	42	156	380	581	816		
	CDD10.0	948	0	0	1	12	74	166	281	265	126	21	2	0		
	CDD18.3	123	0	0	0	0	2	12	49	48	12	0	0	0		
	CDH23.3	2373	0	0	0	9	96	236	847	874	299	12	0	0		
	CDH26.7	878	0	0	0	1	18	63	322	361	113	1	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	360	18	15	20	31	45	81	36	33	33	21	15	15		
	PrecMax	550	48	36	45	69	90	208	126	91	68	69	31	34		
	PrecMin	211	8	3	1	6	1	6	3	1	0	4	2	0		
	PrecStd	81	10	8	12	16	26	41	27	23	20	15	8	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	12.4	17.8	24.5	28.7	32.2	34.5	34.9	32.7	25.0	18.5	11.4		
		MCWB	4.8	6.5	8.0	11.2	13.6	17.5	16.4	16.0	14.9	12.5	9.0	5.1		
	2%	DB	7.3	8.2	14.9	20.5	26.2	28.3	32.4	32.7	30.2	20.9	14.0	7.6		
		MCWB	3.1	4.0	6.8	9.9	13.0	15.5	16.3	15.4	14.6	10.6	7.2	3.5		
	5%	DB	5.3	5.3	12.2	17.2	23.5	25.9	30.2	30.7	26.9	17.8	11.2	5.4		
		MCWB	2.3	2.0	5.7	7.9	11.9	14.8	16.4	15.2	13.4	9.4	5.7	2.0		
	10%	DB	3.8	3.4	9.2	14.7	20.8	23.9	28.2	28.5	23.7	15.0	8.7	3.5		
		MCWB	1.5	0.9	3.9	6.5	11.0	13.8	16.0	15.1	12.2	7.9	4.2	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	6.6	8.9	12.0	14.4	19.2	20.7	18.2	17.0	13.8	9.7	6.0		
		MCDB	9.6	12.0	16.6	21.7	25.9	25.5	27.8	26.7	25.9	23.5	17.4	9.9		
	2%	WB	3.4	4.0	7.4	10.1	13.4	17.4	19.2	17.3	15.3	11.6	7.7	3.6		
		MCDB	6.7	7.7	14.0	19.6	23.6	26.5	26.1	27.0	26.5	18.6	13.1	7.1		
	5%	WB	2.4	2.2	5.9	8.5	12.5	15.9	18.0	16.5	14.1	9.9	6.1	2.2		
		MCDB	5.1	4.9	11.5	16.1	21.4	23.6	25.8	26.8	25.1	16.3	10.9	5.0		
	10%	WB	1.5	1.0	4.2	6.9	11.5	14.7	17.0	15.7	12.9	8.5	4.5	1.0		
		MCDB	3.8	3.3	8.8	13.6	19.2	21.7	25.3	25.8	22.9	14.5	8.3	3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.0	11.5	12.8	13.1	13.1	15.7	16.0	14.9	12.9	11.4	10.3		
		MCDBR	10.6	12.3	15.2	18.3	18.1	17.2	19.6	20.5	20.6	17.1	14.1	11.1		
		MCWBR	7.6	8.5	8.4	9.1	7.4	7.1	7.5	7.2	8.1	8.4	8.2	7.7		
		MCDBR	10.2	11.9	14.6	17.4	15.6	15.4	16.0	16.9	18.9	15.8	13.6	11.2		
	5% WB	MCWBR	7.6	8.6	8.5	9.0	6.8	7.2	7.3	6.5	7.8	8.2	8.3	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.265	0.286	0.335	0.352	0.347	0.347	0.368	0.326	0.291	0.261	0.250		
	taud		2.440	2.483	2.487	2.351	2.346	2.414	2.425	2.338	2.429	2.518	2.505	2.432		
	Ebn at Noon		820	906	938	913	906	912	906	867	876	860	817	783		
	Edh at Noon		61	74	88	113	119	112	109	113	93	71	57	53		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.27	3.56	4.92	5.85	6.53	7.10	5.71	4.07	2.55	1.45	1.01		
	RadStd		0.11	0.19	0.27	0.32	0.44	0.38	0.37	0.36	0.32	0.24	0.10	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page